

# HALEEMA TALLAT

Machine Learning Engineer — Models, Evaluation & Production Inference

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## EXPERIENCE

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### AI Engineer — IgniteTech (via Crossover) Jul 2025 – October 2026 · Remote

- Production LLM systems. Built ingestion and inference infrastructure for an AI customer-support platform handling 1,000+ messages/day across 10–15 messaging channels, routing normalized conversational events to hosted LLM models for NLP and response generation.
- Agent tooling and trace observability. Built internal MCP integrations for coding/engineering agent workflows and added Langfuse instrumentation so teams could trace, replay, and debug multi-turn LLM and tool-call interactions — the telemetry needed to judge whether agent behaviour holds up on real traffic.
- Scalable event processing. Kafka pipelines for up to 27 channel engines with KEDA lag-based autoscaling, Prometheus monitoring, graceful draining; decomposed Java monoliths into stateless Node/TypeScript services on Kubernetes, cutting memory 6–10× and startup ~20×.

### Computer Vision Engineer — Brick & Mortar AI Mar – Oct 2025 · Lahore, PK

- Evaluation integrity and data as the product. Trained a ResNet-12 prototypical-network few-shot classifier on damaged/glared barcode images, then found the reported numbers were an artifact of the harness: split leakage, a 1-shot vs 3–40-shot support mismatch, and 3-way vs 11-way evaluation mismatch. Rebuilt the held-out split and episode sampler; under the corrected 11-way protocol the model reached 89.7% validation accuracy, against 46.7% under the broken setup.
- Shipped to production. Converted PyTorch weights to ONNX and TensorRT, wrote the Triton configuration, and resolved shape-mismatch and inference-runner registration failures in the deployed stack; GPU-accelerated RTSP decoding raised pipeline throughput ~10–15× over naive full-resolution decoding.
- Multi-camera tracking and re-identification. YOLOv8/v11 segmentation → 512-d ReID embeddings from segmented person pixels only → cross-camera association; cut identity switches 20% versus the full-bounding-box baseline via a custom Triton ReID backend with a re-tuned matching threshold.

### Computer Vision Engineer — NextBridge Nov 2024 – Mar 2025 · Lahore, PK

- Real-time WebRTC control. Built the end-to-end WebRTC architecture for a smart wheelchair — signalling server (SDP offer/answer, ICE negotiation), joystick sender, Jetson ROS 2 receiver — with heartbeat/timeout detection, backoff reconnection, a watchdog that force-stops on link loss, and round-trip latency instrumentation.

### Machine Learning Engineer — Powersoft19 Nov 2023 – Nov 2024 · Lahore, PK

- Owned fault/anomaly detection on device telemetry from data to production, replacing manual visual inspection and cutting turnaround from ~48 hours to ~30 minutes.
- GPU honesty. Optimised CUDA/PyTorch training and inference for GPU-bound workloads: mixed-precision training, pinned-memory host-to-device transfers, and channels-last layout for tensor-core utilisation.

### Research Engineer — KICS, Center for Language Engineering, UET Lahore Nov 2022 – Nov 2023

- Read primary literature on retrieval-augmented generation and text encoding and reproduced methods from it; built Tajziyakar, a containerised OCR and segmentation pipeline that extracts and flags TV news ticker content at archive scale.

## EDUCATION

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- **BSc Electrical Engineering, University of Engineering & Technology (UET) Lahore, 2023.** Final-year project (led the group): captured benign and attack traffic from a live IoT testbed rather than using a public benchmark, then compared an LSTM over traffic sequences against Random Forest/XGBoost flow baselines, deployed on Jetson to weigh accuracy against real edge latency and memory. Coursework: linear algebra, probability and statistics, optimisation, signals and systems, DSP.
- **CureMD Master Inventor Award (2022)** for a cancer-treatment-assistant bot for oncologists, built during an AI internship automating oncology guideline processing (~50% less manual effort). **COHACK by DevNation 2022** hackathon winner.

## TECHNICAL SKILLS

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- Core: Python, PyTorch, NumPy, C++, SQL, FastAPI
- Training & evaluation: few-shot and metric learning, prototypical networks, episodic and fixed-episode eval harnesses, paired comparisons, confidence intervals, leakage-aware split design, synthetic episode/support sampling, anomaly detection, MLflow
- LLM inference & agents: vLLM, open-weight model serving, RAG, text encoding, LangChain, MCP agents and tool use, Langfuse tracing, Claude Code/Codex/Cursor agent workflows
- GPUs & deployment: CUDA optimisation, mixed precision, pinned memory, channels-last, ONNX, TensorRT, Triton Inference Server, Jetson, Docker, Kubernetes, Kafka, CI/CD; WebRTC (signalling, SDP/ICE, reconnection, latency instrumentation), RTSP, ROS 2